

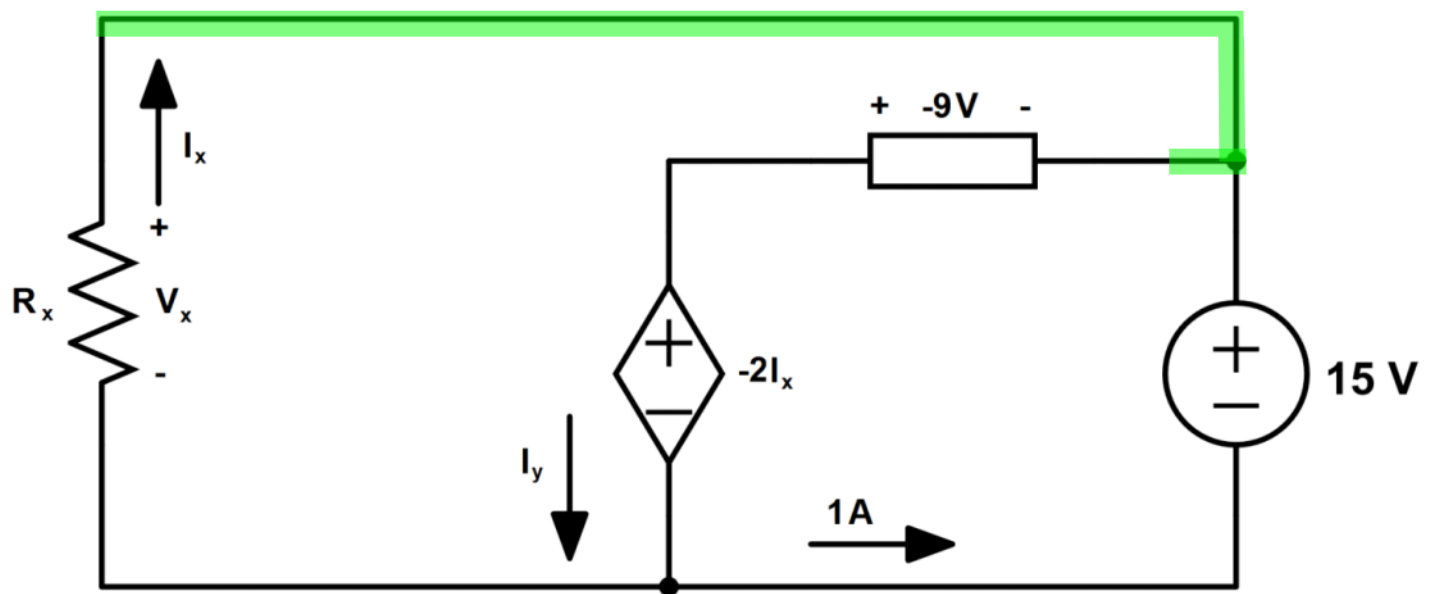
Practice Quiz 1 Solutions

Thursday, August 21, 2025

5:53 PM

For the circuit shown below:

- Find I_y
- Find R_x
- Calculate the power of the unknown element. State whether the power is absorbed or delivered by the unknown element.



KVL Right Loop:

$$-2I_x + 9 - 15 = 0 \rightarrow I_x = -3A$$

We have 2 options for a KCL,
I will choose the highlighted node

$$-I_x - 1 + I_y = 0$$

$$I_y = I_x + 1 \rightarrow I_y = -2A$$

Ohm's Law : $V = IR$

$$V_x = I_x R_x \rightarrow 15 = -(-3) \cdot R_x$$

$$R_x = 5\Omega \quad \text{Resistors are always positive!}$$

Power of the unknown element :

$$P = -IV = -(-9) \cdot (-2) = -18W$$

-18W Delivered